# DHCP: Dynamic Host Configuration Protocol

# IP addresses: how to get one?

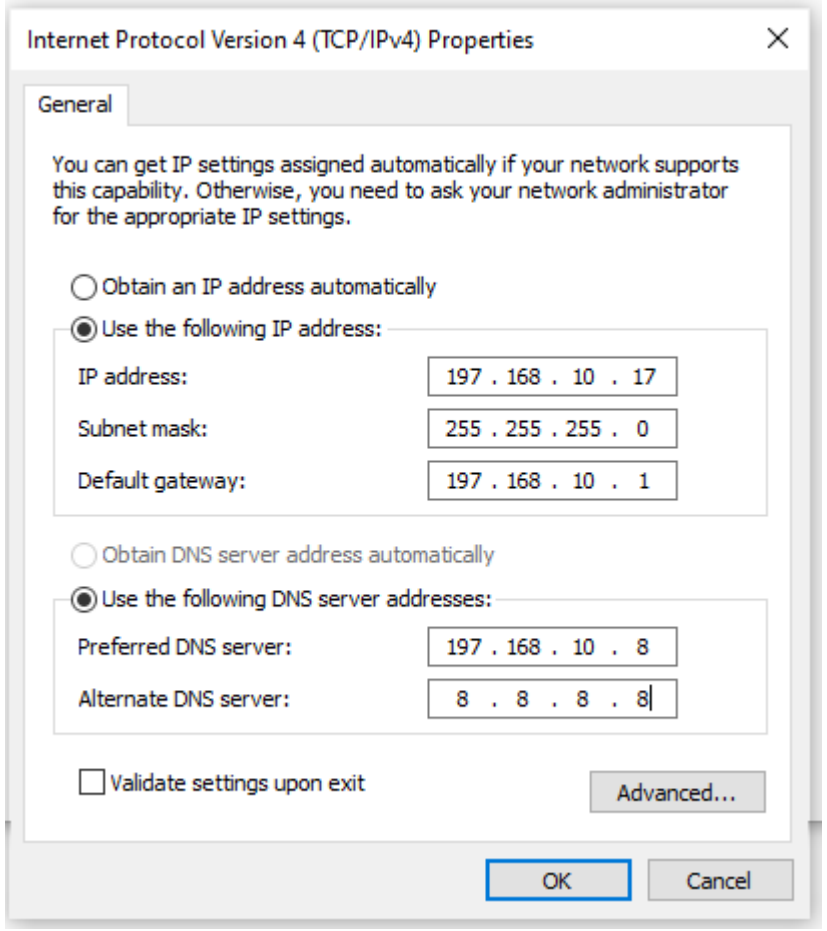
That's actually two questions:

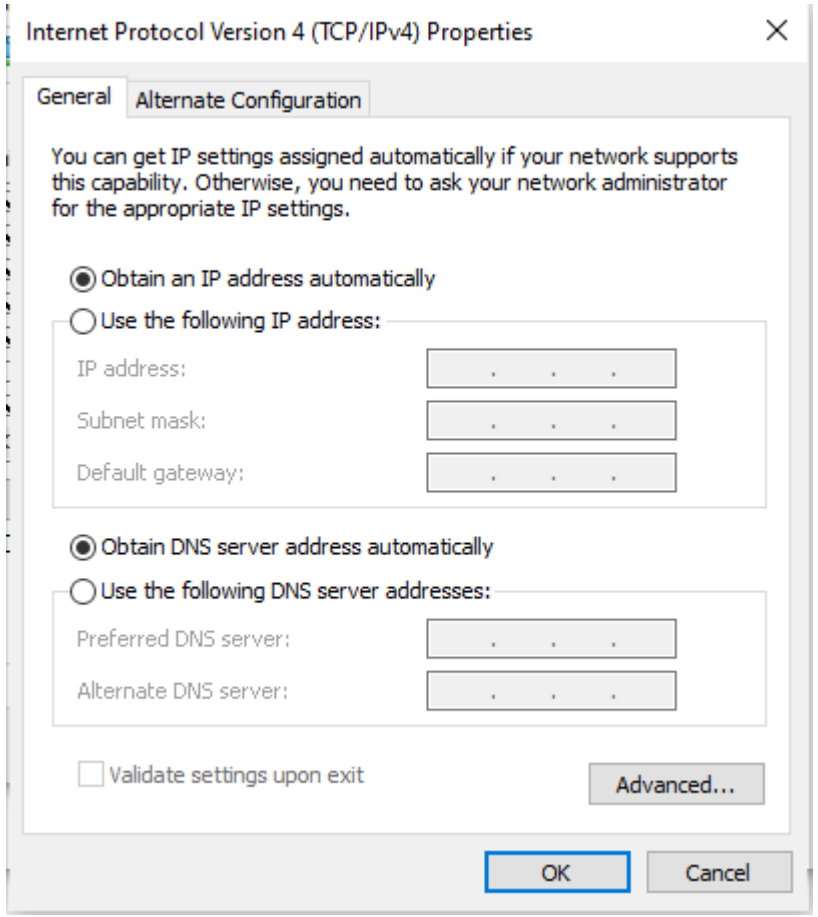
1. Q: How does a *host* get IP address within its network (host part of address)?
2. Q: How does a *network* get IP address for itself (network part of address)

How does *host* get IP address?

- hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
- DHCP: Dynamic Host Configuration Protocol: dynamically get address from as server
  - "plug-and-play"

# IP addresses: how to get one?



# IP Cont..

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : iitbhilai.ac.in
   Description . . . . . . . . . . . : Qualcomm QCA9377 802.11ac Wireless Adapter
   Physical Address. . . . . . . . . : B0-68-E6-82-D9-8D
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::80d1:147e:1fc0:c043%9(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.3.54.107(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.192.0
   Lease Obtained. . . . . . . . . . : 29 September 2020 09:39:45
   Lease Expires . . . . . . . . . . : 29 September 2020 18:28:38
   Default Gateway . . . . . . . . . : 10.3.0.1
   DHCP Server . . . . . . . . . . . : 10.1.72.7
   DHCPv6 IAID . . . . . . . . . . . : 162556134
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-24-F3-49-C3-D8-D0-90-5B-50-B5
   DNS Servers . . . . . . . . . . . : 192.168.10.87
                                       192.168.10.72
```

# DHCP: Dynamic Host Configuration Protocol

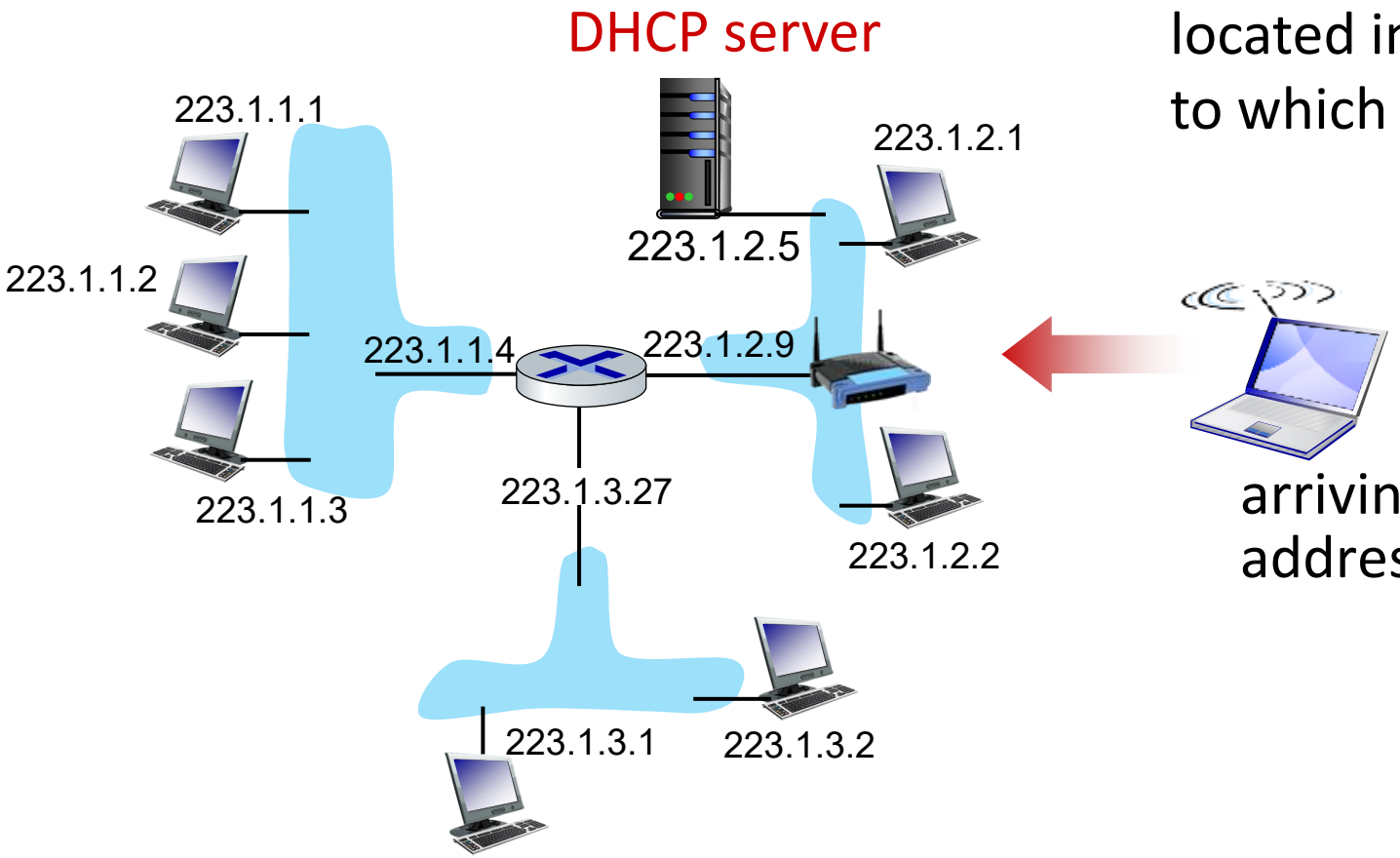
goal: host *dynamically* obtains IP address from network server when it "joins" network

- can renew its lease on address in use
- allows reuse of addresses (only hold address while connected/on)
- support for mobile users who join/leave network

DHCP overview:

- host broadcasts DHCP discover msg [optional]
- DHCP server responds with DHCP offer msg [optional]
- host requests IP address: DHCP request msg
- DHCP server sends address: DHCP ack msg

# DHCP client-server scenario

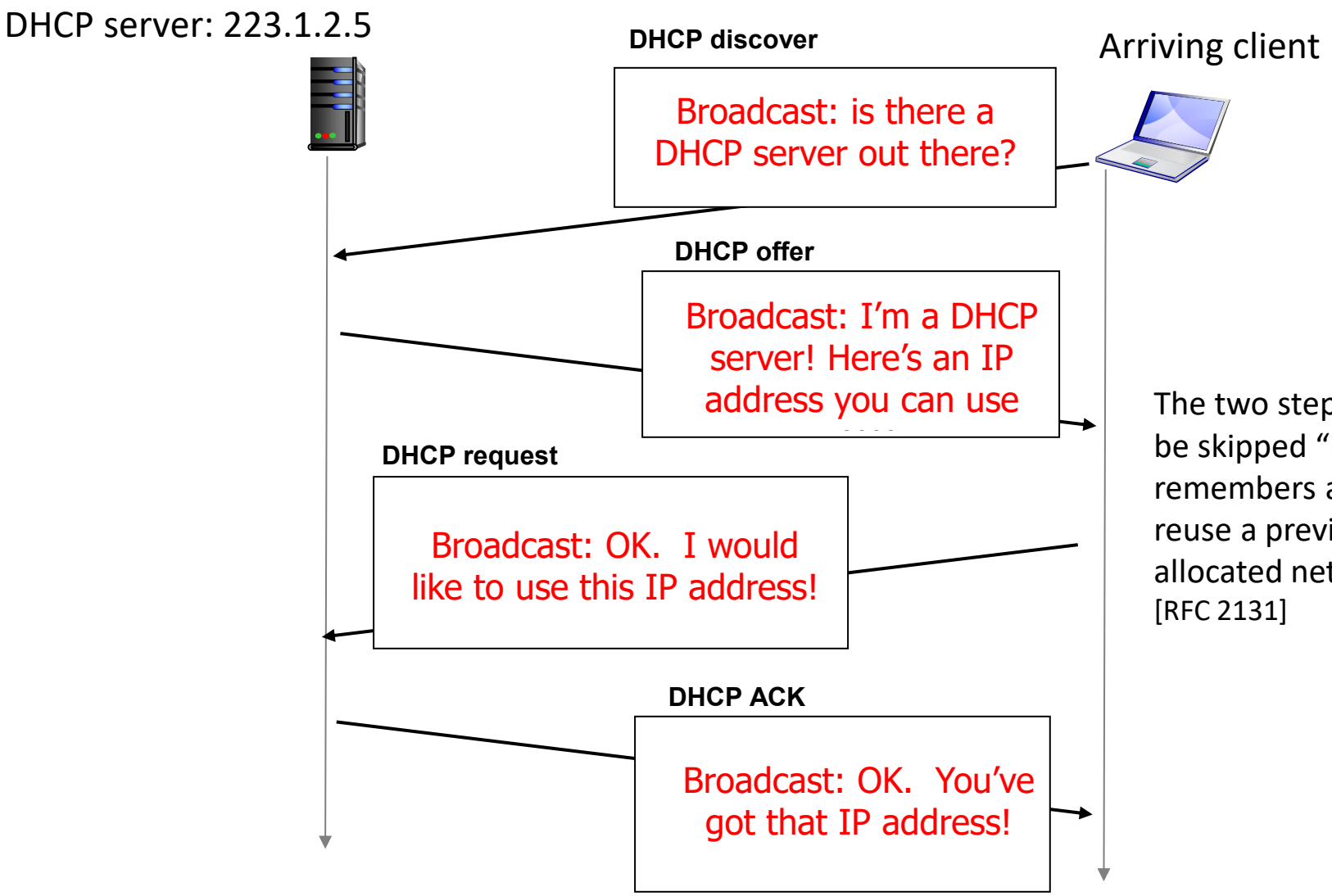



Typically, DHCP server will be co-located in router, serving all subnets to which router is attached



arriving DHCP client needs address in this network

# DHCP client-server scenario



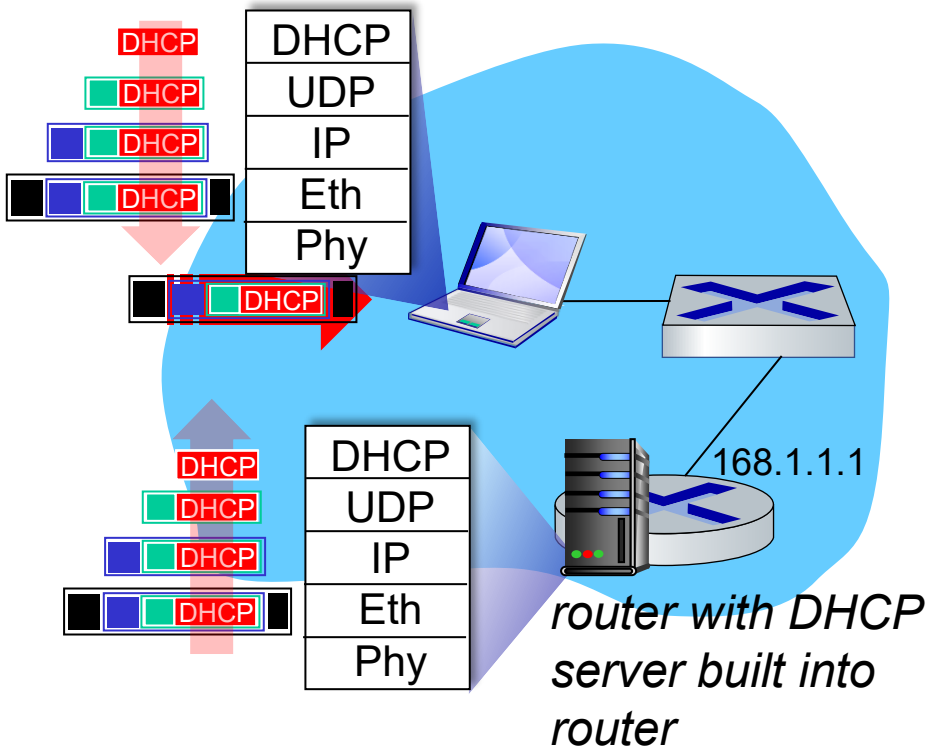

The two steps above can be skipped "if a client remembers and wishes to reuse a previously allocated network address" [RFC 2131]

# DHCP: more than IP addresses

DHCP can return more than just allocated IP address on subnet:

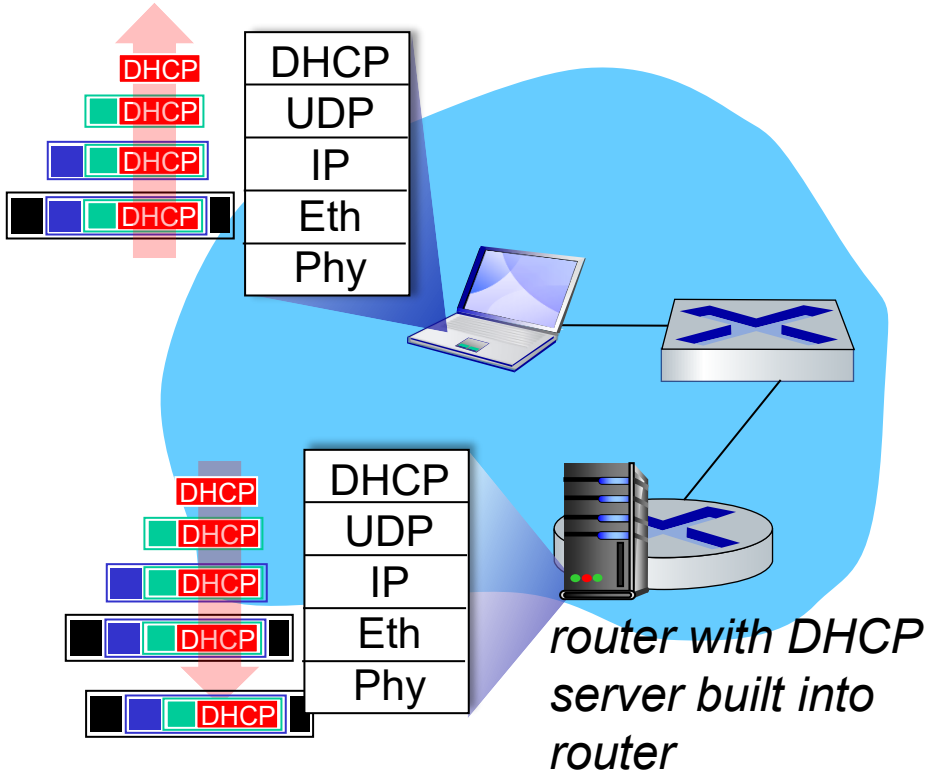
- address of first-hop router for client
- name and IP address of DNS sever
- network mask (indicating network versus host portion of address)

# DHCP: example



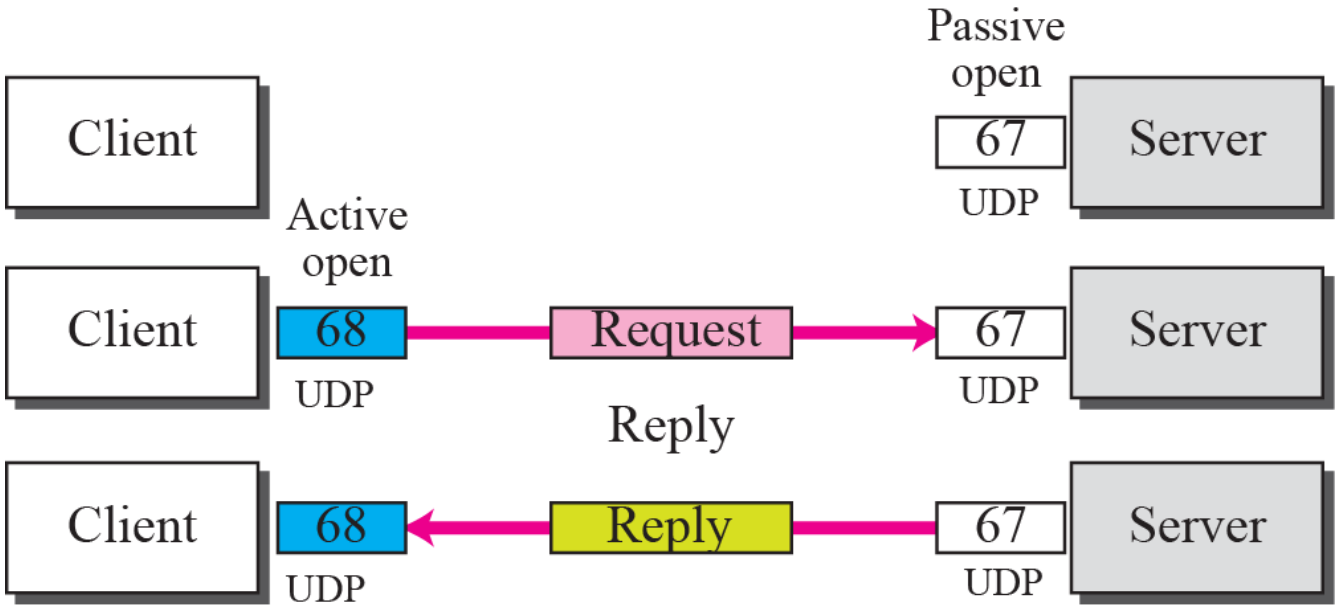

- Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server.
- DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- Ethernet frame broadcast (dest: FFFFFFFFFFFF) on LAN, received at router running DHCP server
- Ethernet demux'ed to IP demux'ed, UDP demux'ed to DHCP

# DHCP: example




- DCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulated DHCP server reply forwarded to client, demuxing up to DHCP at client
- client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

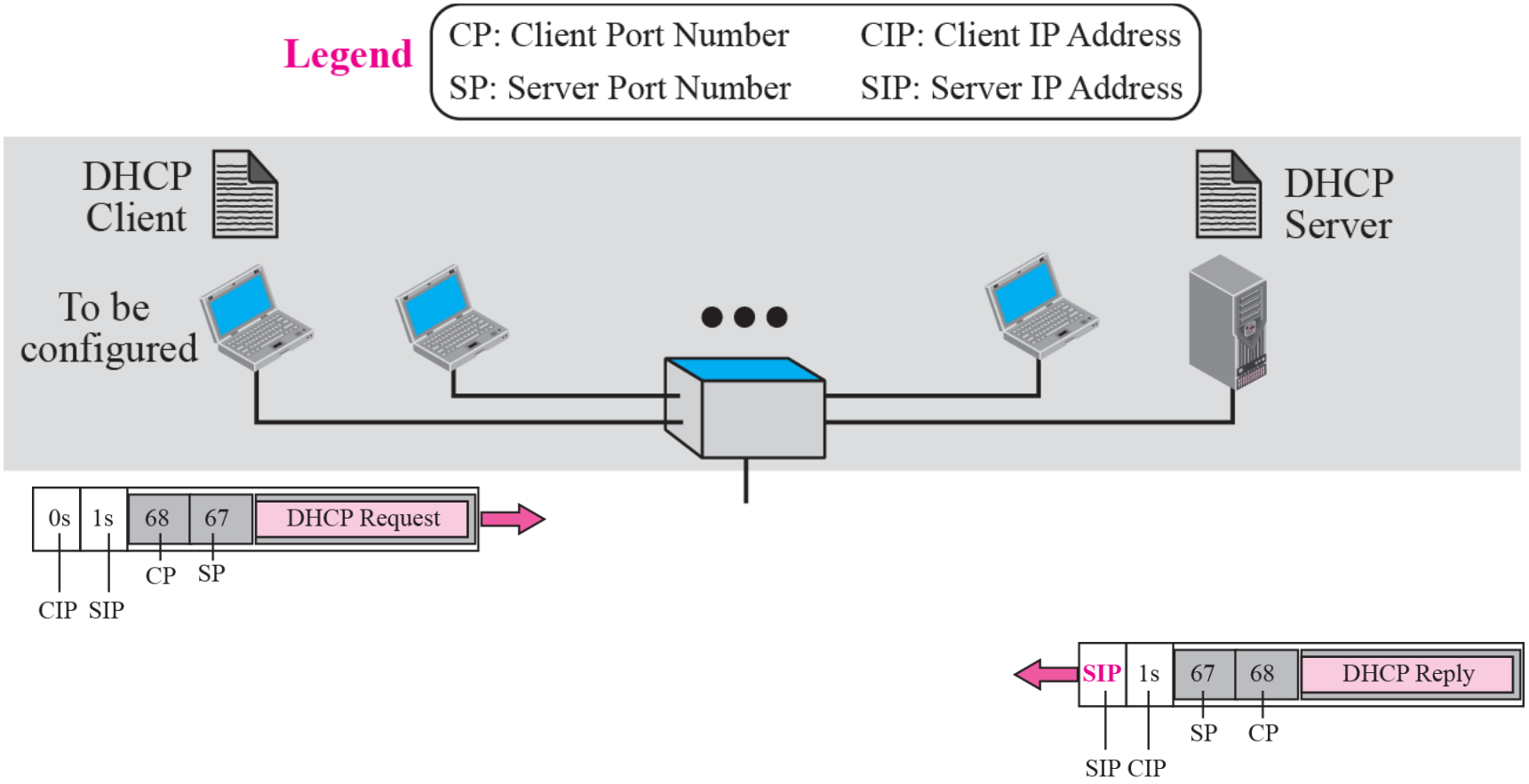
*Use of UDP ports*



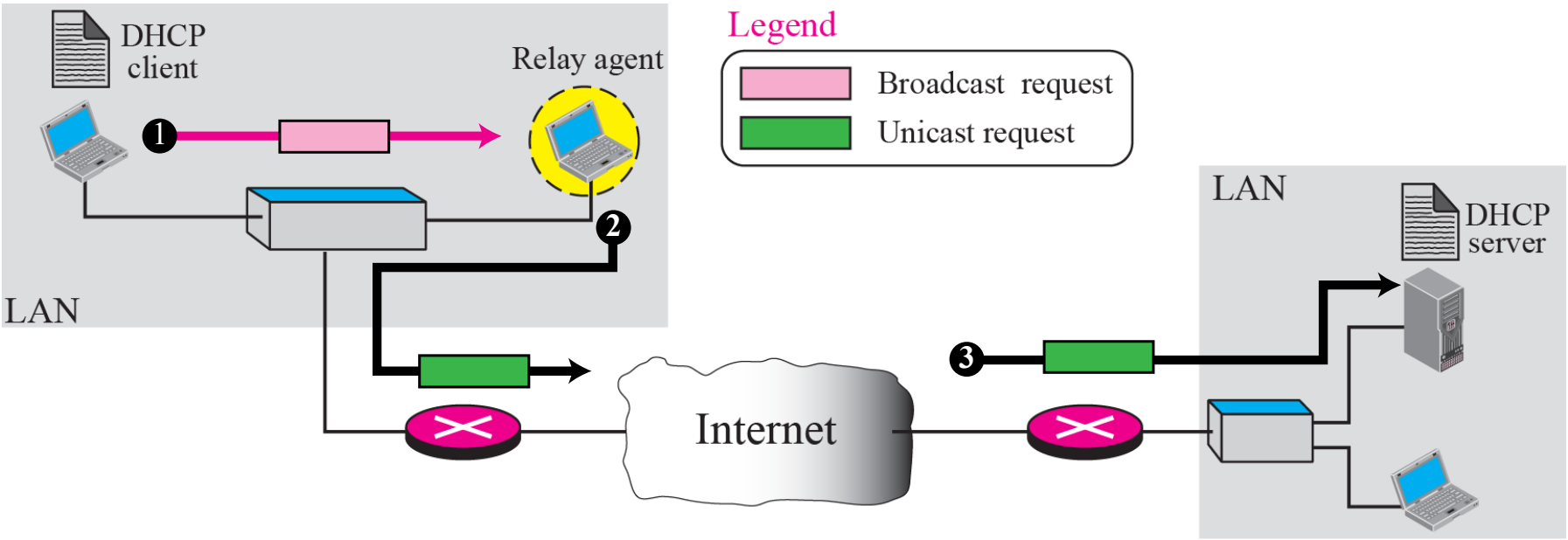
# DHCP Operation

The DHCP client and server can either be on the same network or on different networks. Let us discuss each situation separately.

*Client and server on the same network*

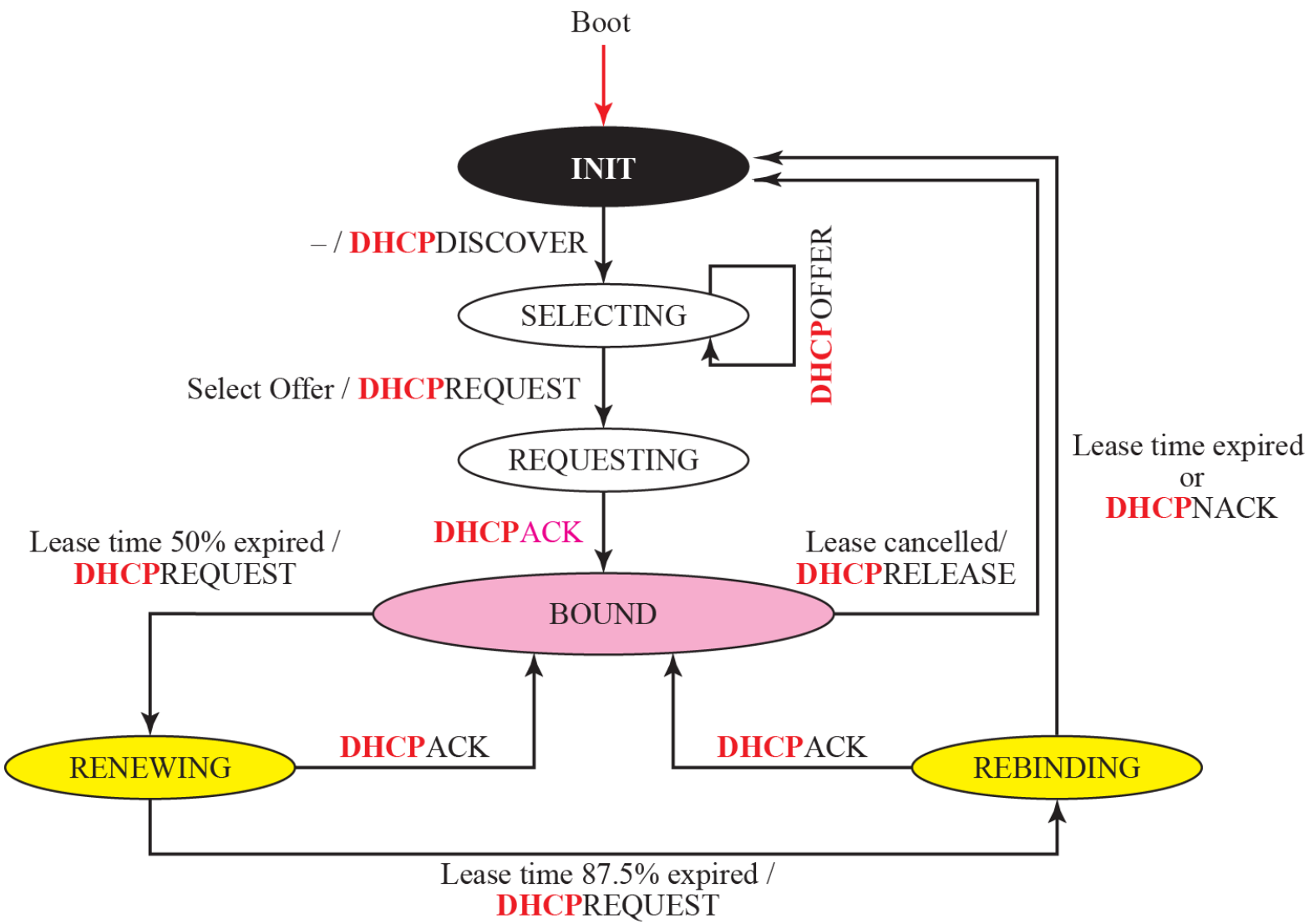


*Client and server on two different networks*

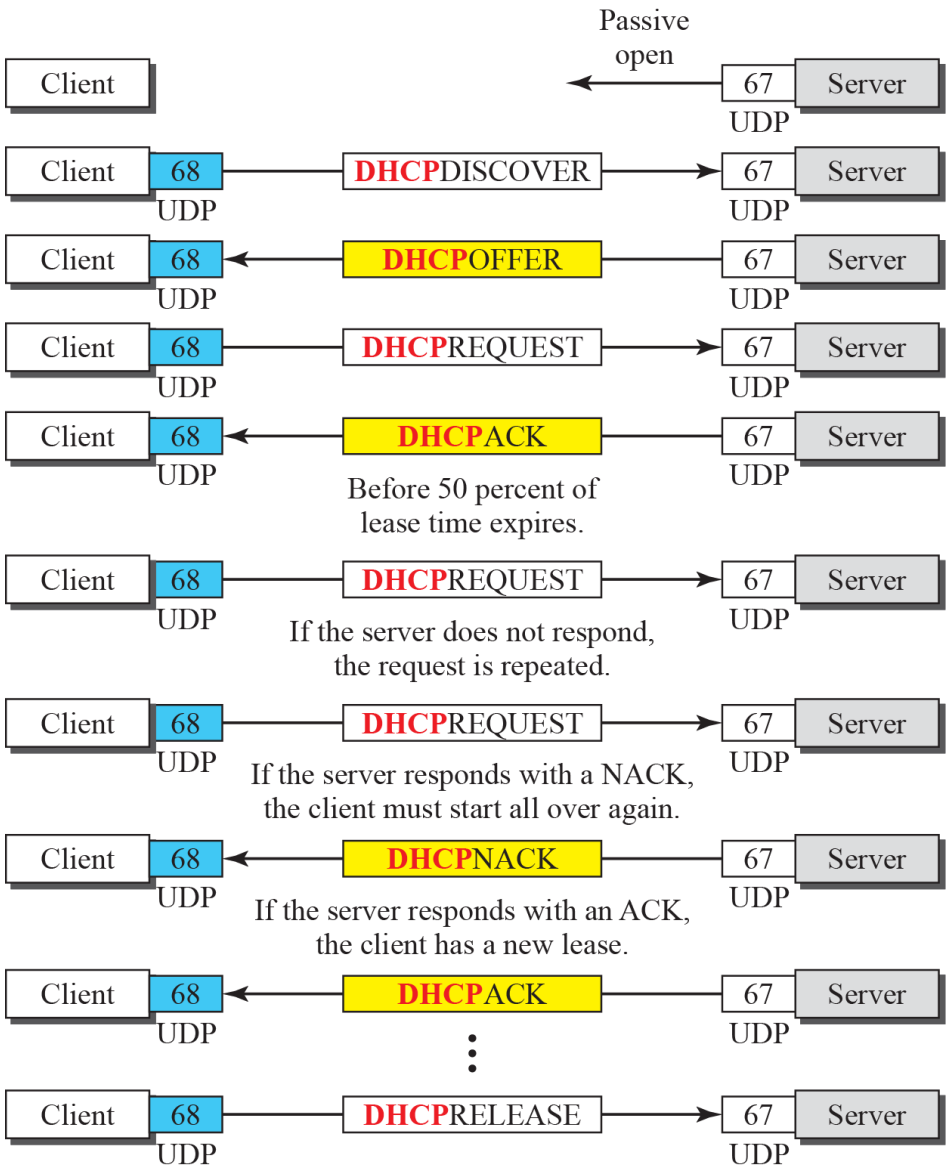



Usually, Routers/layer 3 switches (default gateway of the subnet) act as relay agent

*DHCP client transition diagram*



# *Exchanging messages*



# IP addresses: how to get one?

*Q:* how does *network* get subnet part of IP address?

*A:* gets allocated portion of its provider ISP's address space

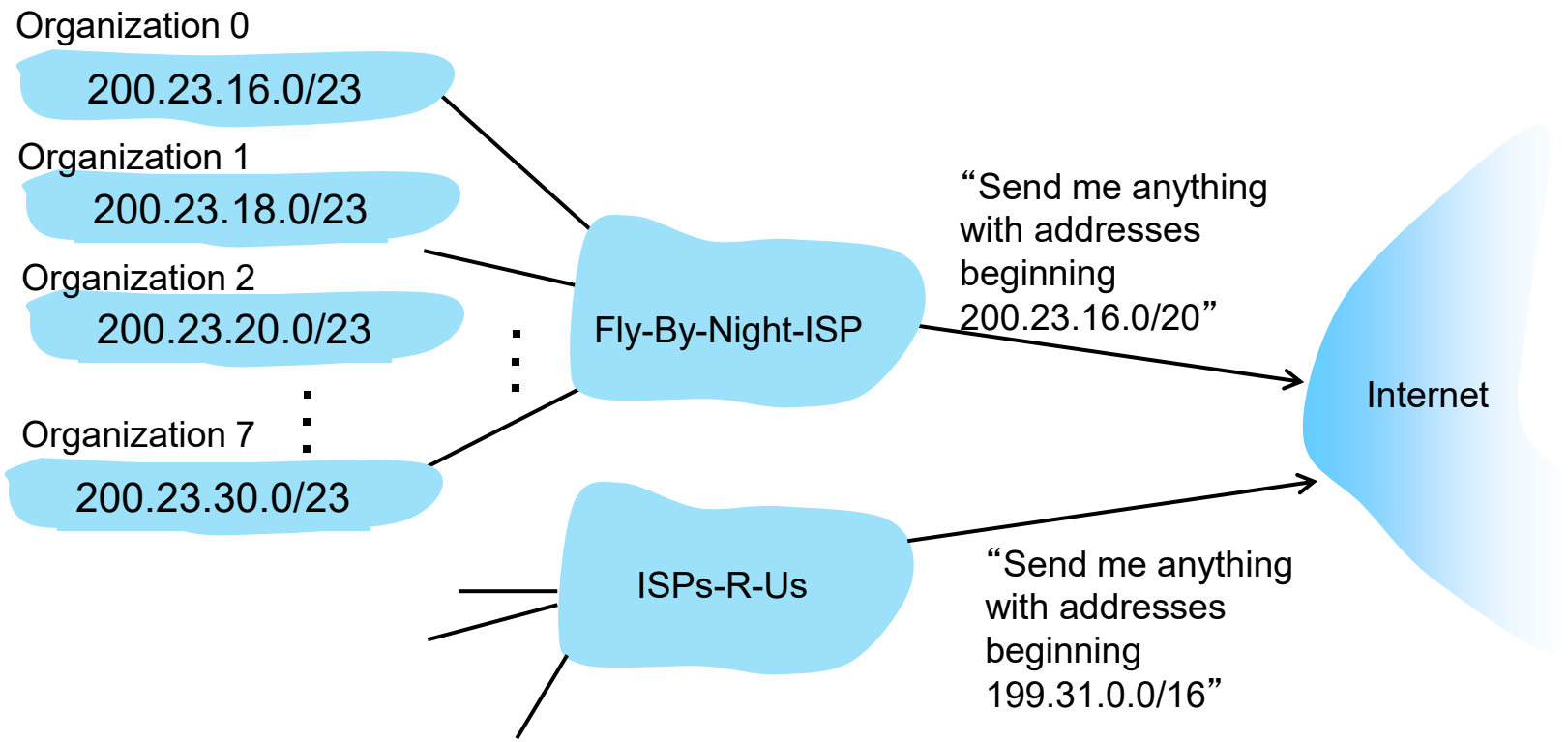
| | | | |
|---|---|---|---|
| ISP's block | 11001000 00010111 00010000 | 00000000 | 200.23.16.0/20 |

ISP can then allocate out its address space in 8 blocks:

| | | | |
|---|---|---|---|
| Organization 0 | 11001000 00010111 00010000 | 00000000 | 200.23.16.0/23 |
| Organization 1 | 11001000 00010111 00010010 | 00000000 | 200.23.18.0/23 |
| Organization 2 | 11001000 00010111 00010100 | 00000000 | 200.23.20.0/23 |
| … | ….. | …. | …. |
| Organization 7 | 11001000 00010111 00011110 | 00000000 | 200.23.30.0/23 |

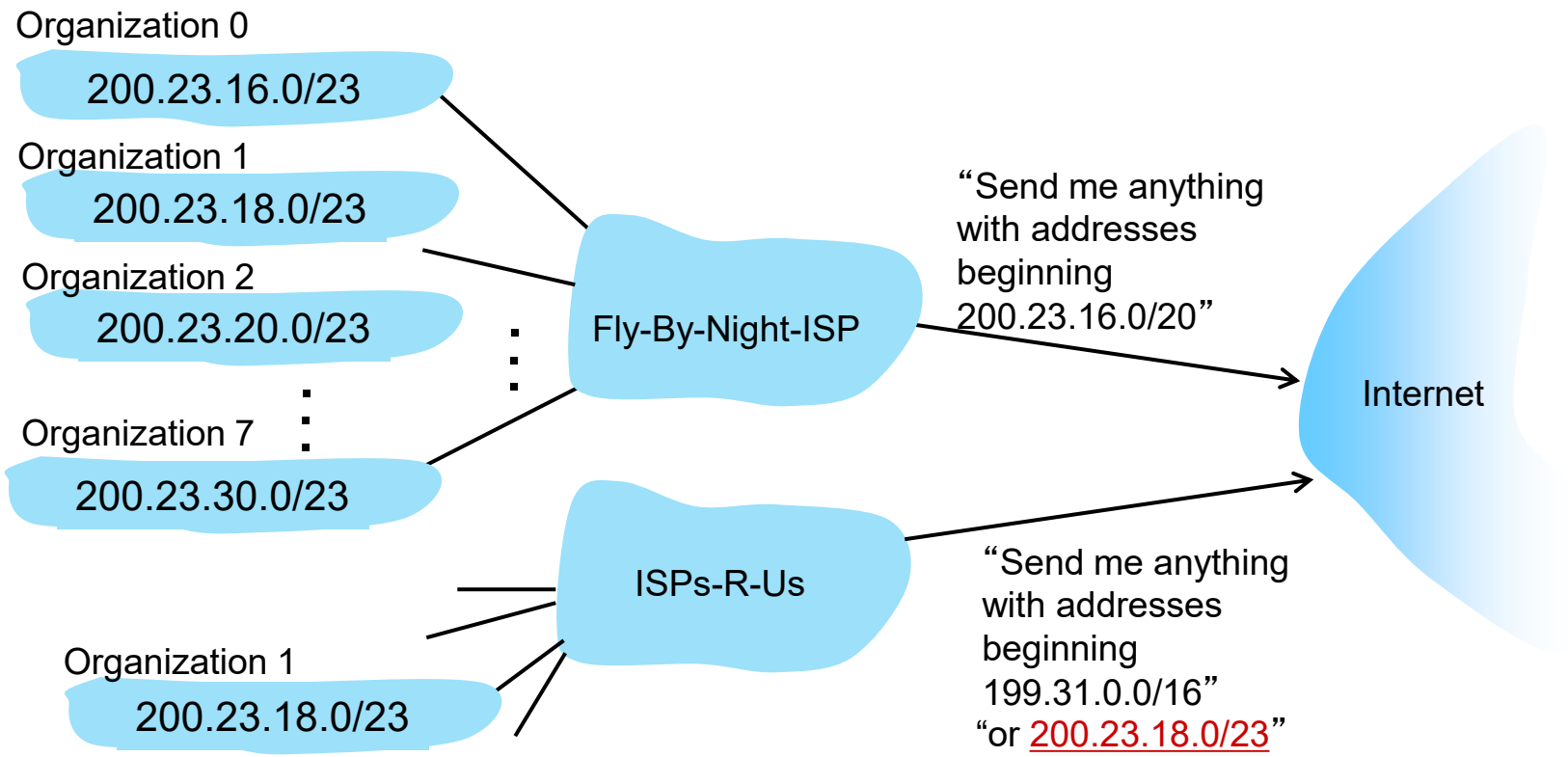
# Hierarchical addressing: route aggregation

hierarchical addressing allows efficient advertisement of routing information:



# Hierarchical addressing: more specific routes

- Organization 1 moves from Fly-By-Night-ISP to ISPs-R-Us
- ISPs-R-Us now advertises a more specific route to Organization 1



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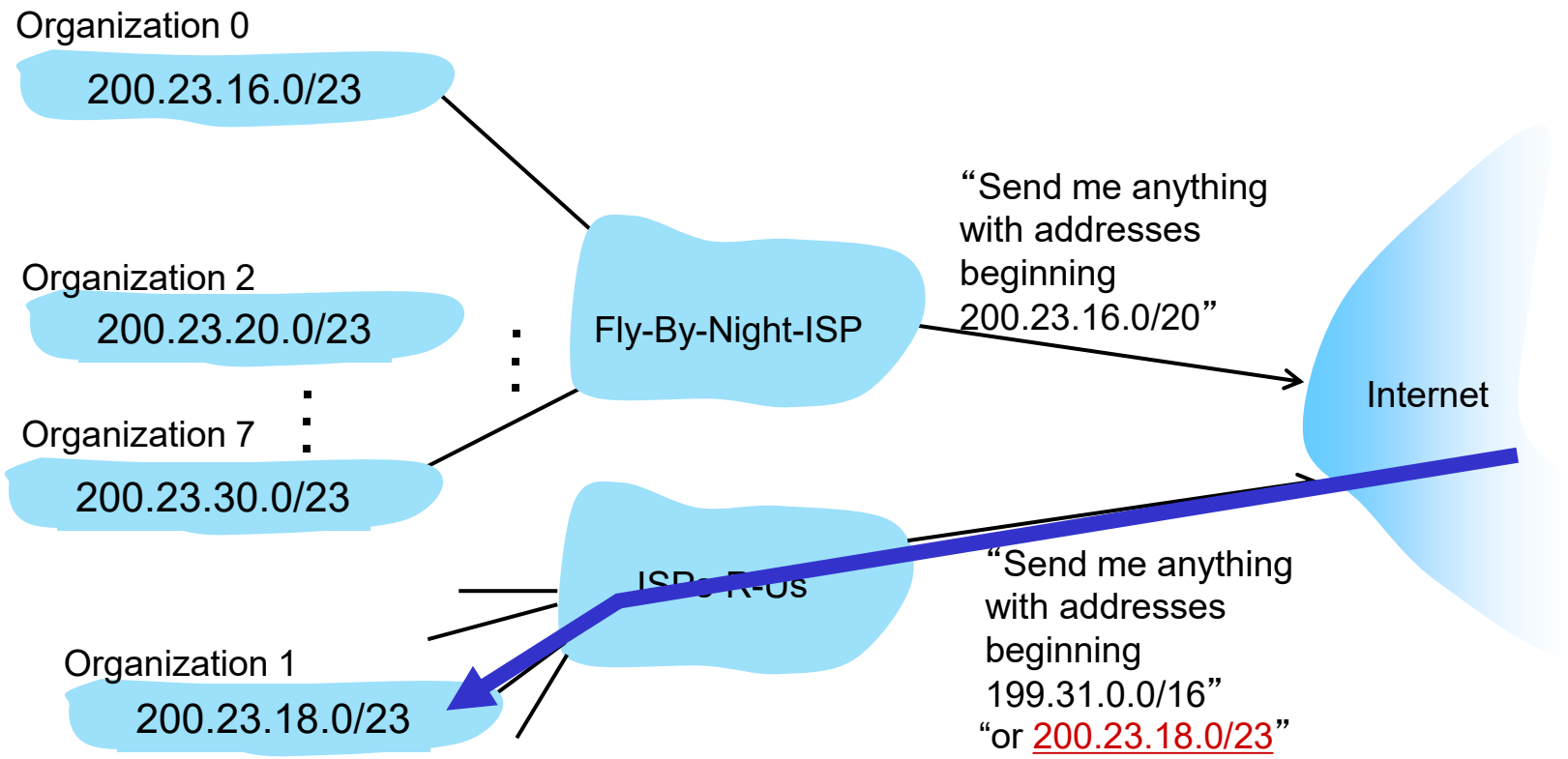


# IP addressing: last words …

*Q:* how does an ISP get block of addresses?

*A:* ICANN: Internet Corporation for Assigned Names and Numbers http://www.icann.org/

- allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- manages DNS root zone, including delegation of individual TLD (.com, .edu , …) management

*Q:* are there enough 32-bit IP addresses?

- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT (next) helps IPv4 address space exhaustion
- IPv6 has 128-bit address space

Public Internet addresses are regulated by five Regional Internet Registries (RIRs):

- ARIN
- RIPE
- APNIC
- LACNIC
- AfriNIC

# NAT: Network Address Translation

# Private Network

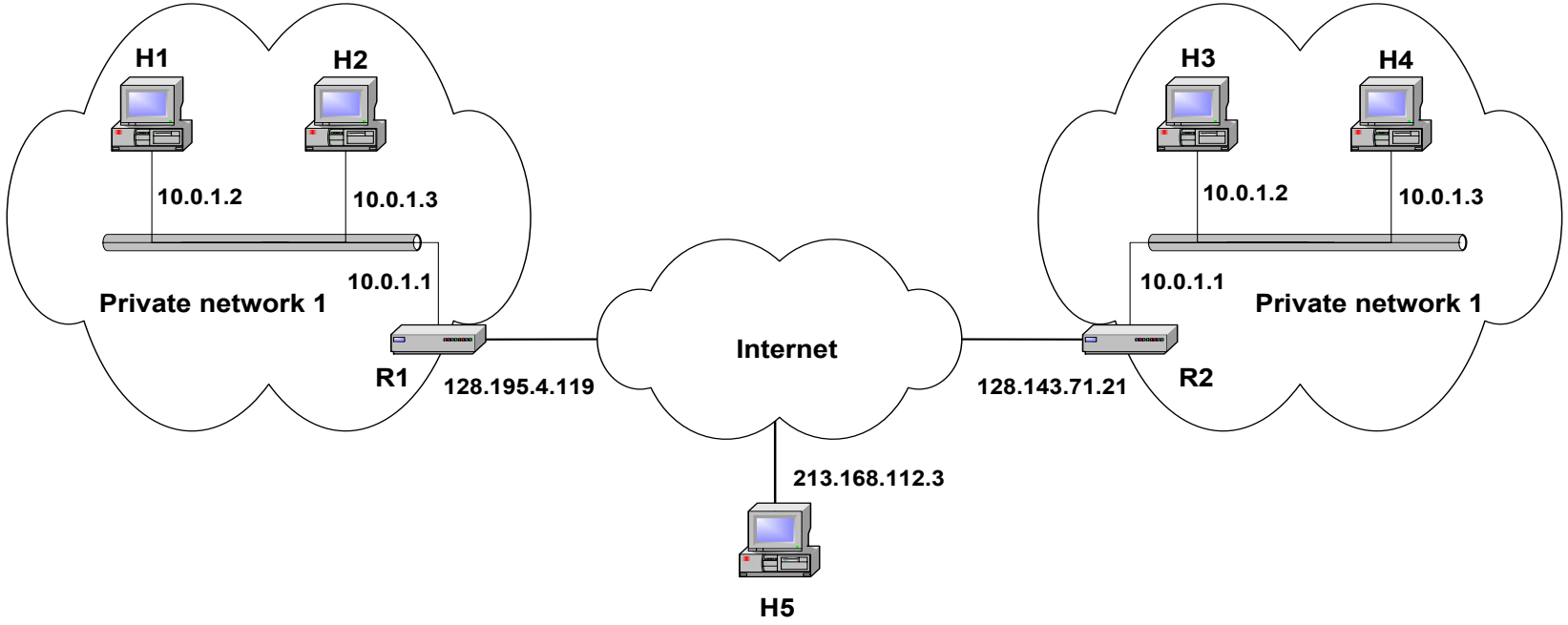
- *Private IP* network is an IP network that is not directly connected to the Internet

- IP addresses in a private network can be assigned arbitrarily.
  - Not registered and not guaranteed to be globally unique

- Generally, private networks use addresses from the following experimental address ranges (**Free and *non-routable addresses***): **[RFC 1918]**
  - 10.0.0.0 – 10.255.255.255
  - 172.16.0.0 – 172.31.255.255
  - 192.168.0.0 – 192.168.255.255

| Class | RFC 1918 Internal Address Range | CIDR Prefix |
|---|---|---|
| A | 10.0.0.0 - 10.255.255.255 | 10.0.0.0/8 |
| B | 172.16.0.0 - 172.31.255.255 | 172.16.0.0/12 |
| C | 192.168.0.0 - 192.168.255.255 | 192.168.0.0/16 |

# Private Addresses



# *Network Address Translation (NAT)*

- RFC 1631

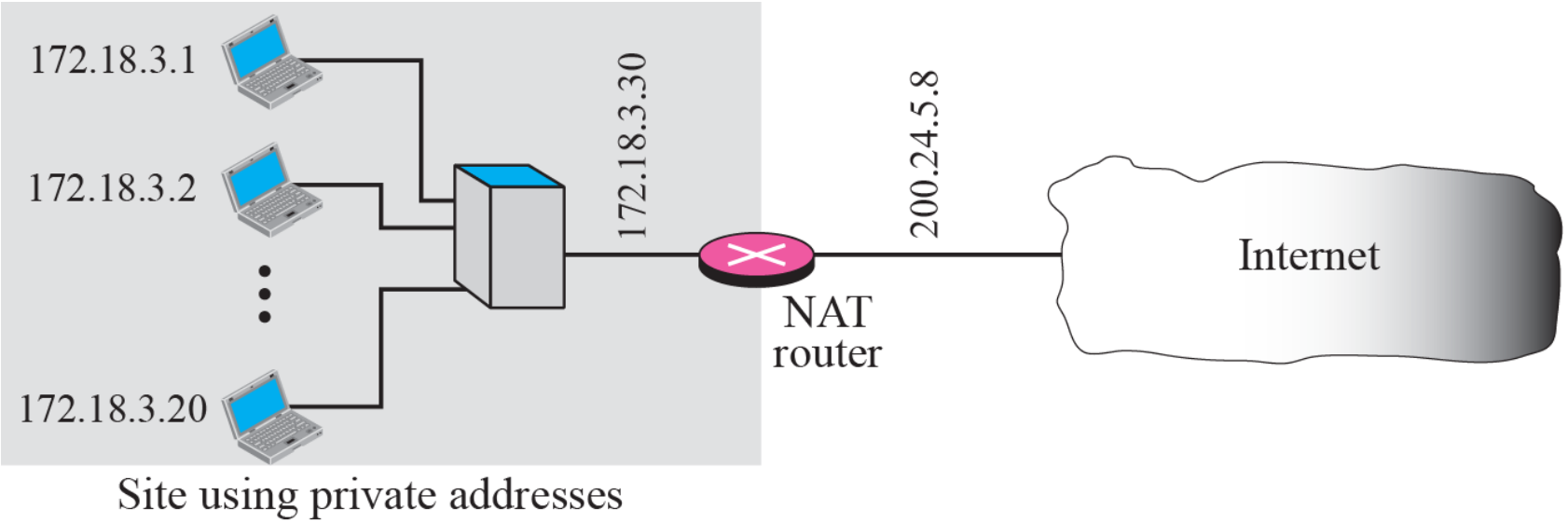
- A short term solution to the problem of the depletion of IP addresses
  - Long term solution is IP v6
  - CIDR (Classless Inter Domain Routing ) is a possible short term solution
  - NAT is another

- NAT is a way to conserve IP addresses
  - Can be used to hide a number of hosts behind a single IP address
  - Uses private addresses:
    - 10.0.0.0-10.255.255.255,
    - 172.16.0.0-172.32.255.255 or
    - 192.168.0.0-192.168.255.255

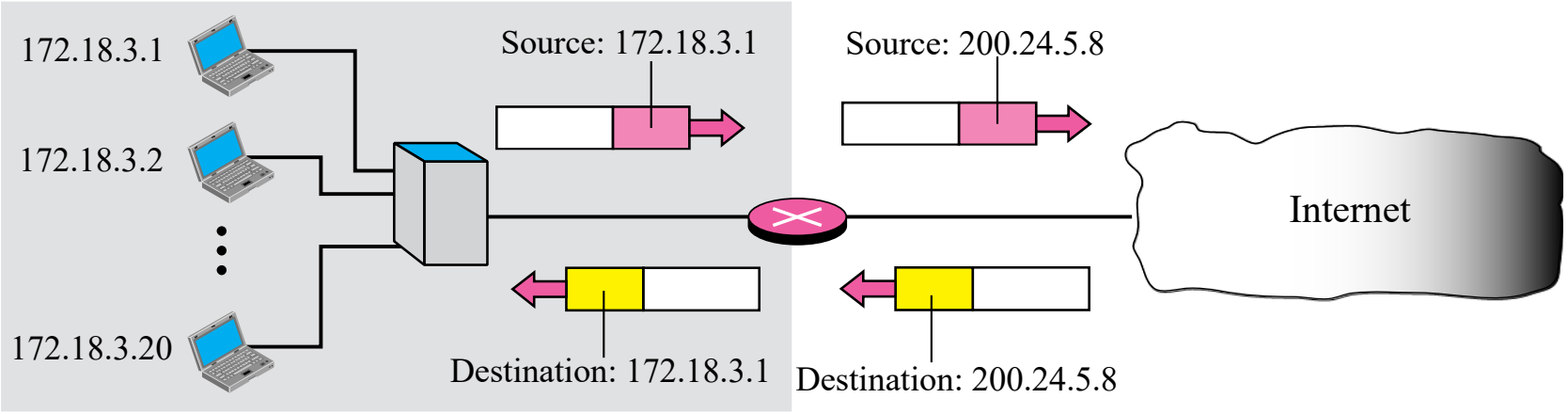
# *Network Address Translation (NAT)*

- NAT is a router function where IP addresses (and possibly port numbers) of IP datagrams are replaced at the boundary of a private network
- NAT is a method that enables hosts on private networks to communicate with hosts on the Internet
- NAT is run on routers that connect private networks to the public Internet, to replace the IP address-port pair of an IP packet with another IP address-port pair.
- NAT translates non-routable, private, internal addresses into routable, public addresses.
- NAT has an added benefit of adding a degree of privacy and security to a network because it hides internal IP addresses from outside networks.

# Network Address Translation (NAT)

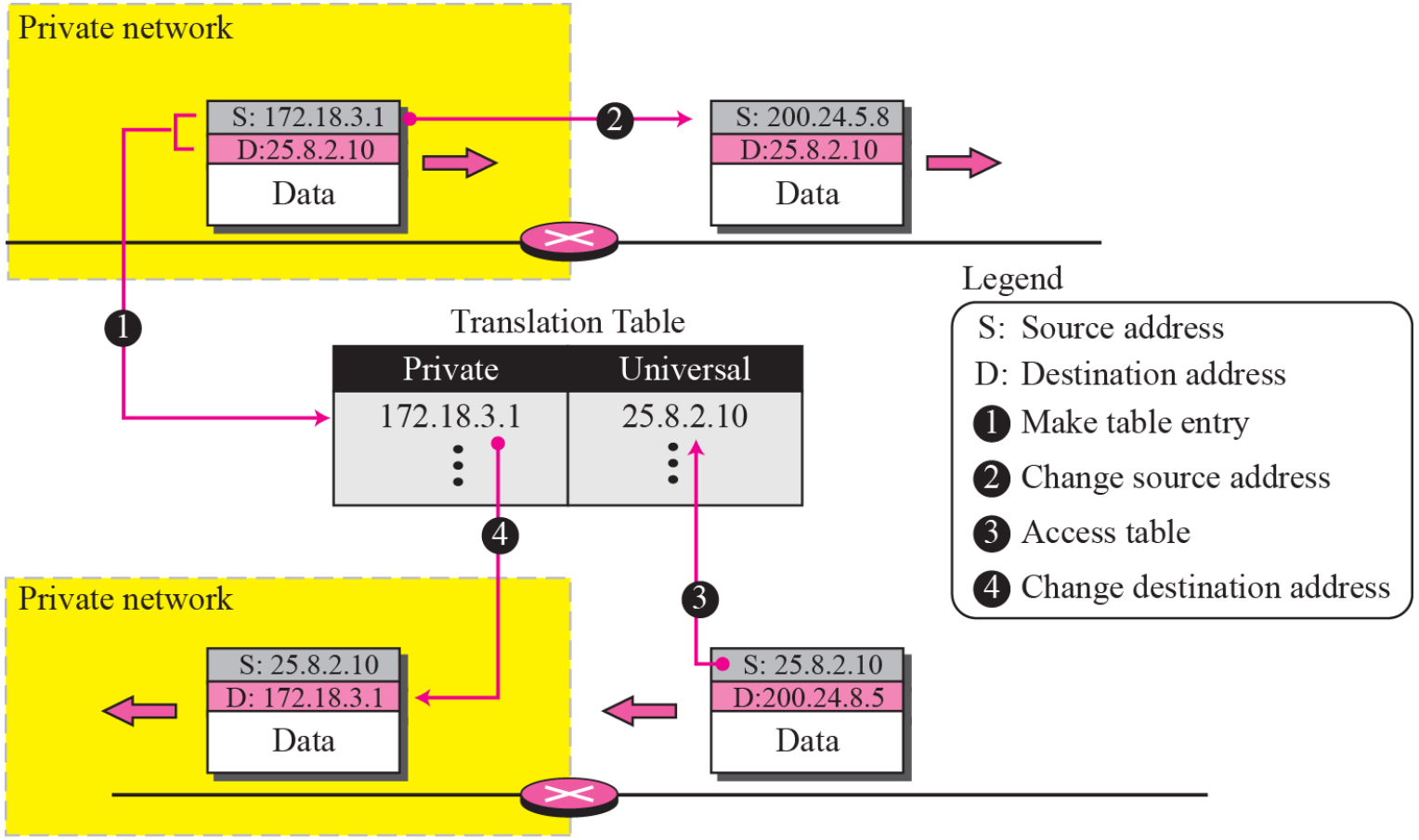


# Address Resolution

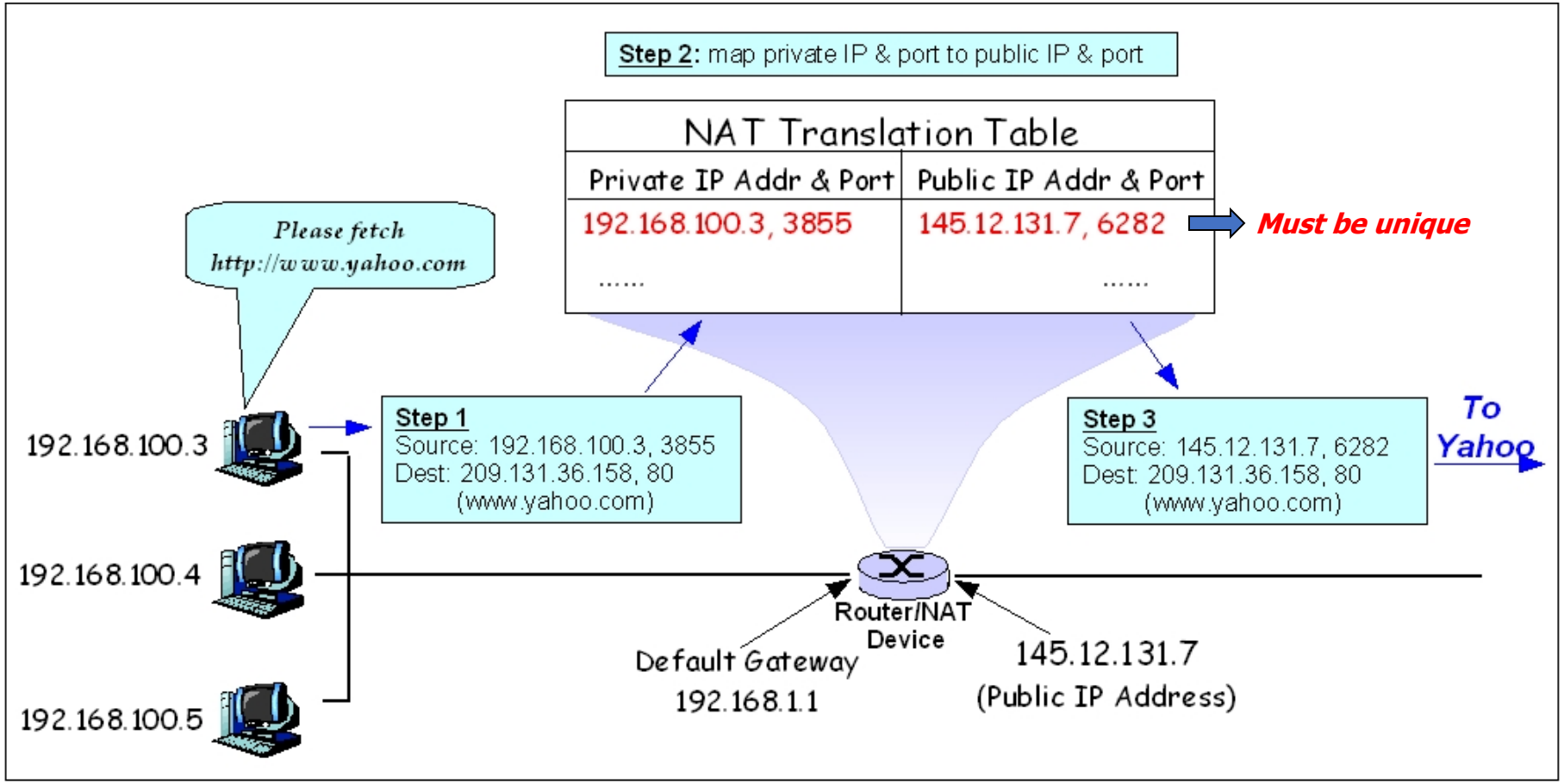



Site using private addresses

# Translation



# NAT Table with IP address & Port #



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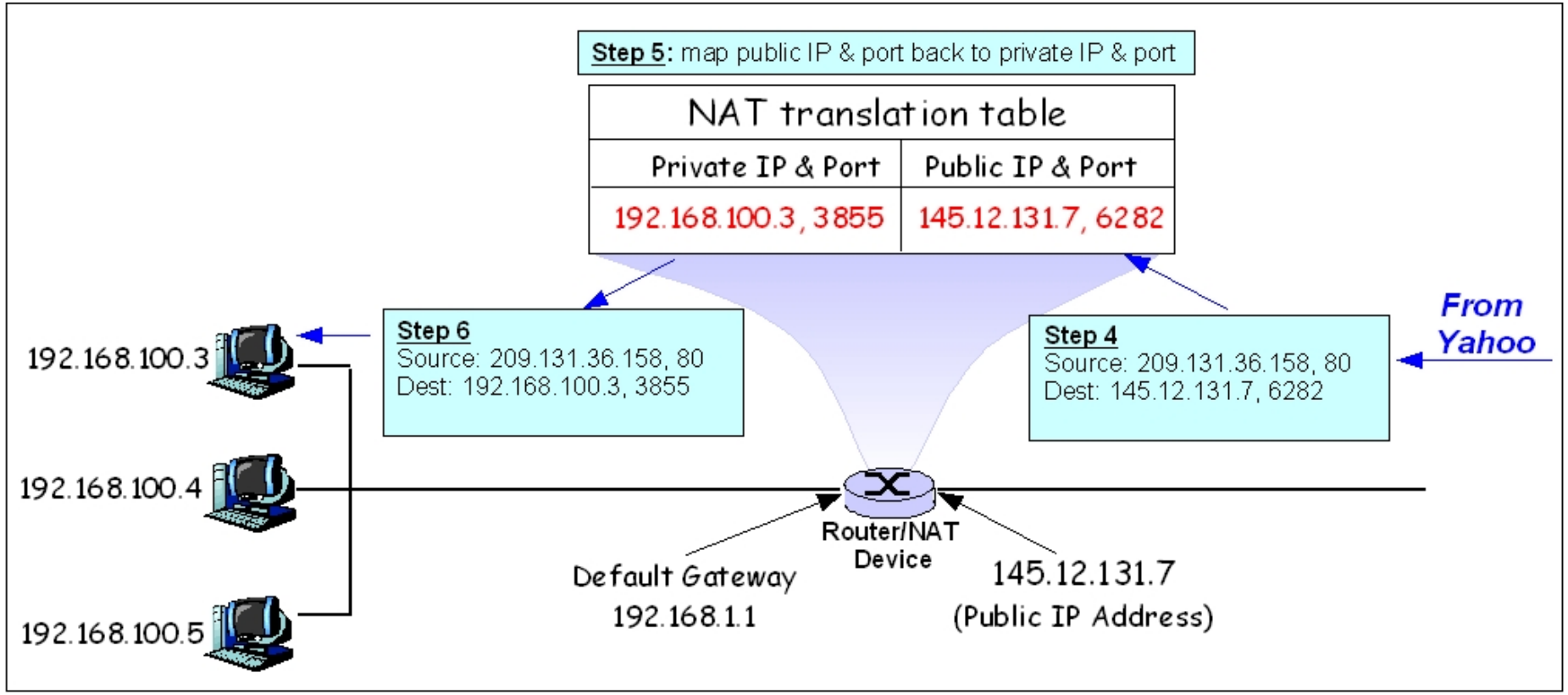

# Concerns about NAT

- **Performance:**
  - Modifying the IP header by changing the IP address requires that NAT boxes recalculate the IP header checksum
  - Modifying port number requires that NAT boxes recalculate TCP checksum

- **End-to-end connectivity:**
  - NAT destroys universal end-to-end reachability of hosts on the Internet.
  - A host in the public Internet often cannot initiate communication to a host in a private network.
  - The problem is worse, when two hosts that are in a private network need to communicate with each other.